

PHISHING EMAIL INVESTIGATION REPORT

Real-World Phishing Email Analysis Project

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Project Objective

To investigate real phishing and suspicious emails through technical analysis including sender verification, authentication checks, email header analysis, malicious link inspection, and threat assessment using industry-standard cybersecurity tools.

Tools Used

- Mozilla Thunderbird
- Visual Studio Code
- VirusTotal
- Email Header Analysis
- Manual IOC Investigation

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EXECUTIVE SUMMARY

This investigation project involved the analysis of three real-world suspicious email samples collected from publicly available phishing repositories. The purpose of this project was to simulate the workflow of a Security Operations Center (SOC) analyst by performing practical phishing email investigations using industry-standard cybersecurity tools.

Each email sample was analyzed through sender verification, email header inspection, SPF/DKIM/DMARC authentication checks, link investigation, and behavioral threat assessment. The investigation identified multiple indicators of compromise, including spoofed sender identities, failed authentication mechanisms, suspicious reply-to domains, tracking URLs, and social engineering tactics.

The findings confirmed that all analyzed email samples demonstrated characteristics commonly associated with phishing campaigns, spam abuse, or credential harvesting attempts.

This project demonstrates practical hands-on cybersecurity skills relevant to SOC operations, phishing analysis, digital forensics investigation, and incident response workflows.

INVESTIGATION METHODOLOGY

The phishing email investigation was conducted using a structured incident analysis methodology similar to real-world SOC investigation workflows.

Step 1: Email Sample Collection

Three suspicious email samples in .EML format were collected from public phishing repositories for practical investigation.

Step 2: Email Header Analysis

Raw email headers were inspected using Visual Studio Code to identify sender details, return-path anomalies, SPF, DKIM, DMARC authentication failures, and suspicious routing paths.

Step 3: Sender Verification

The displayed sender address, actual sending domain, reply-to address, and mail relay infrastructure were compared to detect spoofing attempts.

Step 4: Link and IOC Analysis

Embedded hyperlinks were extracted from email bodies and analyzed using VirusTotal to identify suspicious redirections, tracking domains, and malicious indicators.

Step 5: Behavioral Threat Assessment

Email content was reviewed for social engineering techniques such as urgency, impersonation, reward baiting, credential theft attempts, and suspicious call-to-action elements.

Step 6: Threat Classification

Each email sample was categorized based on observed attack characteristics such as phishing, spam abuse, impersonation, or credential harvesting attempts.

EMAIL SAMPLE 1 ANALYSIS — BANK PHISHING CAMPAIGN

Email Sample Overview

Sample 1 was identified as a phishing email impersonating Banco Bradesco, a financial institution, using reward-based social engineering tactics to create urgency and manipulate the recipient into clicking a malicious link.

Technical Findings

Displayed Sender:

banco.bradesco@atendimento.com.br

Subject:

CLIENTE PRIME - BRADESCO LIVELO: Seu cartão tem 92.990 pontos LIVELO expirando hoje!

Authentication Results:

SPF: TempError

DKIM: None

DMARC: TempError

CompAuth: Fail

Suspicious Indicators Identified

- Sender domain does not match legitimate banking infrastructure
- Missing DKIM signature
- Failed authentication checks
- Suspicious reward baiting subject line
- External tracking / redirect URL detected
- Social engineering urgency tactic (“points expiring today”)

Threat Assessment

This sample demonstrates characteristics of a credential phishing campaign designed to impersonate a banking rewards notification to lure victims into clicking a malicious redirection link.

EMAIL SAMPLE 2 ANALYSIS — MICROSOFT IMPERSONATION PHISHING

Email Sample Overview

Sample 2 was identified as a phishing email impersonating Microsoft account security alerts to create fear and urgency regarding unusual sign-in activity.

Technical Findings

Displayed Sender:

Microsoft account team <no-reply@access-accsecurity.com>

Reply-To Address:

sotrecognizd@gmail.com

Subject:

Microsoft account unusual sign-in activity

Authentication Results:

SPF: None

DKIM: None

DMARC: PermError

CompAuth: Fail

Suspicious Indicators Identified

- Fake Microsoft impersonation domain
- Reply-to address redirects to Gmail
- Failed SPF authentication
- Missing DKIM signature
- DMARC validation failure
- Fear-based social engineering tactic
- Fake security alert impersonation

Threat Assessment

This sample demonstrates a classic credential phishing campaign impersonating Microsoft security notifications to steal user credentials through fear-based deception.

EMAIL SAMPLE 3 ANALYSIS — SPAM / MALICIOUS ADVERTISING CAMPAIGN

Email Sample Overview

Sample 3 was identified as a suspicious spam-based email campaign promoting solar panel offers through mass unsolicited email distribution and suspicious redirection infrastructure.

Technical Findings

Displayed Sender:

zonnepaneel@appjj.serenitepure.fr

Reply-To Address:

news@aichakandisha.com

Subject:

Zonnepanelen voor een goede prijs

Authentication Results:

SPF: None

DKIM: None

DMARC: None

CompAuth: Fail

Suspicious Indicators Identified

- Unrelated sender and reply-to domains
- Failed authentication validation
- Missing DKIM signature
- Marketing spam redirection infrastructure
- Suspicious tracking URL
- Social engineering advertising lure
- Possible mass phishing / spam abuse campaign

Threat Assessment

This sample demonstrates characteristics of spam abuse infrastructure potentially used for malicious advertising, phishing redirection, or mass unsolicited email distribution.

OVERALL FINDINGS

The investigation of three suspicious email samples revealed multiple indicators commonly associated with phishing, impersonation, and spam-based malicious campaigns.

Key Findings

- 100% of analyzed email samples failed one or more authentication security checks (SPF, DKIM, DMARC, or CompAuth)
- Multiple sender spoofing and impersonation techniques were identified
- Social engineering tactics included urgency, fear, fake security alerts, reward baiting, and deceptive advertising
- Suspicious redirect / tracking URLs were observed in multiple samples
- Reply-To address mismatches were detected, indicating potential credential harvesting attempts
- Legitimate brand impersonation was used to manipulate user trust (Banking and Microsoft)
- One sample demonstrated characteristics of mass unsolicited spam infrastructure with suspicious promotional redirection

Threat Classification Summary

Sample 1:

Credential Phishing / Banking Impersonation

Sample 2:

Credential Phishing / Microsoft Security Impersonation

Sample 3:

Spam Abuse / Suspicious Advertising / Possible Malicious Redirection

Risk Assessment

Based on observed indicators, the analyzed samples represent medium to high risk threats capable of credential theft, phishing redirection, user deception, and potential malware delivery.

CONCLUSION

This phishing email investigation project successfully analyzed three real-world suspicious email samples using practical cybersecurity investigation techniques aligned with SOC and digital forensics workflows.

The investigation identified multiple indicators of compromise, including sender spoofing, failed authentication mechanisms (SPF, DKIM, DMARC, and CompAuth), suspicious redirect infrastructure, deceptive reply-to addresses, brand impersonation, and social engineering tactics designed to manipulate user behavior.

Each analyzed sample demonstrated distinct malicious characteristics, ranging from credential phishing and impersonation attacks to spam-based suspicious advertising campaigns. The findings confirmed that even emails appearing visually legitimate can contain significant security threats when examined through technical analysis.

This project strengthened practical skills in email header analysis, phishing detection, IOC investigation, threat classification, and cybersecurity reporting using industry-relevant tools such as Mozilla Thunderbird, Visual Studio Code, and VirusTotal.

Overall, this investigation demonstrates the importance of technical email analysis in identifying phishing threats and highlights the critical role of cybersecurity analysts in protecting users and organizations from email-based attacks.